

Star Health

Star Health AI Tutor

A Proof of Concept (POC) for voice-to-voice AI tutoring

v1.0 — POC · Sept 2026

Submitted By



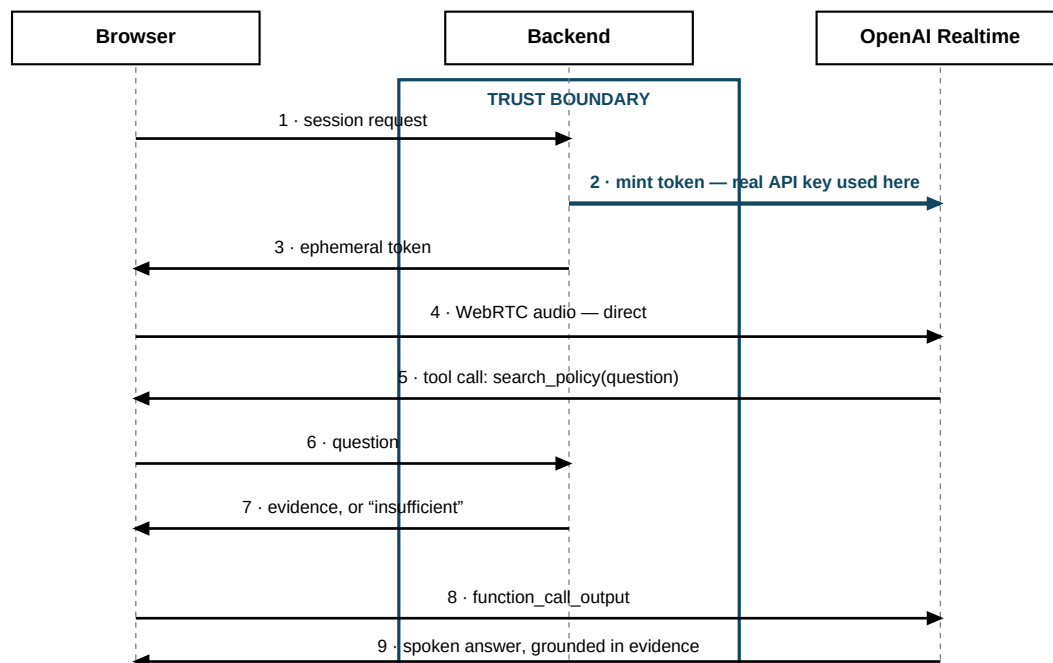
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Goal

A browser app where one speech-to-speech model explains a single health-insurance policy document out loud, lets the user interrupt at any point with a question, and answers every policy question strictly from that document — never from general knowledge, and never a guess.

Functional Architecture



The real API key (blue boundary) never leaves the backend. Audio runs straight between browser and model for low latency (step 4); every policy question is forced through the tool round-trip (steps 5–9) so the model can only answer from retrieved evidence.

Design Principles

- **One realtime model, no pipeline stitching.** Speech-to-text, reasoning, text-to-speech, and interruption handling are one model over one WebRTC connection — not three services glued together.
- **Evidence before answer.** The system prompt forces a search_policy tool call for any policy-specific question; the model is never allowed to answer from general insurance knowledge.
- **A fixed refusal, not a guess.** When retrieval confidence falls below threshold, the model returns one exact sentence — never an improvised, possibly wrong, answer.

- **Credentials and data stay server-side.** The browser only ever holds a short-lived session token and plain evidence text — never the API key, never the vector index.
- **The retriever is swappable.** One interface, `PolicyRetriever.search(question)`, isolates today's local JSON store from tomorrow's hosted vector search — nothing else in the system needs to change.

Solution Approach

Concern	Choice	Why
Voice model	OpenAI Realtime API (gpt-realtime) over WebRTC	Single model covers STT, reasoning, TTS, and barge-in — no LiveKit, no Pipecat, no multi-agent orchestration. Preferred over Gemini Live API because it connects browser-to-model over native WebRTC, versus Gemini's WebSocket transport, which needs a hand-built raw-PCM audio pipeline in the browser.
Auth	Backend mints a short-lived ephemeral token per session	The real API key is used exactly once per session, server-side, and never reaches the browser.
Tool execution	Browser relays the tool call; backend does the retrieval	This is how Realtime function-calling works — but the vector index and embeddings never leave the backend.
RAG store	Local JSON file, brute-force cosine similarity	Simplest reliable option for one policy document at POC scale — no database server, no ops.
Ingestion	One-off script: PDF -> chunks -> embeddings -> index	Runs once at setup; keeps the runtime path free of PDF parsing.

Security & Guardrails

- Browser holds only an ephemeral Realtime token — never the actual API key.
- Every policy question is forced through `search_policy`; the model cannot see or query the vector store directly.
- Answers are constrained to retrieved evidence by the system prompt, with one fixed fallback sentence when evidence is insufficient.